



Old Bridge Township School District

Curriculum & Instruction Department

Course: 7th Grade Honors Math

Department: Math

Term: Semester

Board Approval	Committee Members	Supervisor	Notes
9/24/2024	Fran Capone	Stacy Winters	
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MISSION STATEMENT

Recognizing that each person has an inherent worth, it is the mission of the Old Bridge Public Schools to empower each student to learn to achieve success and to become a confident, productive member of a pluralistic, global society prepared to face the challenges of the 21st Century.

BELIEFS

- Learning is a lifelong process.
- All children can learn.
- Education is a shared responsibility among the schools, the child, the family and the community.
- All students are entitled to a safe, supportive learning environment.
- Self-esteem is necessary for successful learning.
- The development of curriculum is an on-going and cyclical process.
- The district should provide a dynamic curriculum that recognizes the various needs and learning styles of our students.
- An appreciation of the arts and humanities enhances the education of the whole child.
- All students should be taught to appreciate and respect the cultural differences in our society.
- All students are worthy of educational opportunity, respect, tolerance, and fairness.
- Schools should prepare students to function successfully in society.
- All children should be encouraged to rise to their highest level of achievement.

AFFIRMATIVE ACTION STATEMENT

The Old Bridge Township Public Schools provides equal educational opportunities for all students, including equal access to all school facilities, programs, equipment, staff services, financial resources, courses, or activities, and other benefits regardless of race, color, creed, religion, sex, ancestry, national origin, social, economic or academic status or physical handicap.

PHILOSOPHY

Old Bridge Township Schools are committed to providing a quality middle school mathematics curriculum that reflects the goals and ideas of the National Council Teachers of Mathematics (NCTM) and is aligned with the New Jersey Student Learning Standards (NJSLS). To achieve this goal, the middle school mathematics curriculum is based on the ideas that mathematics is best learned through active participation through activities that integrate both conceptual and algorithmic thinking in mathematics. We believe students benefit from seeing mathematical ideas presented in a real-world context where they can transfer their knowledge of skills to solve problems. This application of their learning also creates a richer and more challenging curriculum. In addition, we strongly feel that the responsibility for student achievement should be a solid partnership - one shared by teacher, parent, and student. With the availability of modern technology, students will be able to delve more deeply into meaningful mathematics using a graphing calculator or other mathematical tools. They will use mathematical modeling to make abstract concepts seem more concrete. Students will finish the year believing and understanding the utility and value of mathematics.

PREFACE

The primary goal of this middle school mathematics program is to create an educational experience that results in children who are flexible mathematical thinkers and who see the connections between the mathematics that they learn in school and the thinking that they do in real life. In addition, mathematics should be a challenging yet enjoyable experience. We believe that the children who learn in this program will benefit from a particular focus on problem solving that is presented in the context of real-world problems. Students should be able to see the connections to real life situations that appear in many of the problems presented here. This vision is guided by the eight mathematical process standards:

1. Make sense of problems and persevere in solving them. Solving problems is a central focus of this curriculum. Students should be able to apply their math skills to solve complex, multi-step problems across all disciplines within mathematics. Working both individually and cooperatively, students will continue to develop number sense skills, proportionality, and statistical analysis through problem solving. In addition, students will use models and apply problem solving processes to solve complex problems.
2. Reason abstractly and quantitatively. As a result of experiencing this curriculum, students should be able to begin to increase the level of their abstract thought by using concrete models to investigate concepts within the program and use that knowledge to begin to make the transition to algebraic thought. Students will explore negative numbers, absolute value, and variables and apply the same to real world occurrences in a mathematical context.
3. Construct viable arguments and critique the reasoning of others. A student will be able to apply the knowledge that he or she gains in this program to begin to make inferences regarding mathematical properties and proofs. Students will work cooperatively to reason and work to solve problems and provide explanations based in an understanding of mathematical concepts. Students should be able to use various reasoning skills using algebraic models such as expressions, equations, graphs, and tables to communicate their understanding of mathematics.
4. Model with mathematics. Students will learn to use models to express solutions to various types of mathematical processes ranging from operations with rational numbers to locating real numbers on a number line to solving equations to displaying data in line plots, box plots, and Venn diagrams. Students will also use variables as a means to represent real life situations and coordinate grids to represent direct and inverse variation.

Marking Period 1		Marking Period 2	
Topic/Unit (use assessments as breaks if you divide a Unit) Identify what content will comprise each major assessment.		Topic/Unit Topic/Unit (use assessments as breaks if you divide a Unit) Identify what content will comprise each major assessment.	
1	Topic 1.1, 1.2, 1.6 – 1.9 – Understand Rational, Multiply/ Divide Integers and Rational Numbers	1	Topic 5 – Solve Problems Using Equations and Inequalities
2	Topic 1.3 – 1.5 – Add/Subtract Integers and Rational Numbers	2	Topic 2 – Analyze and Use Proportional Relationships
3	Topic 4 – Generate Equivalent Expressions	3	Topic 3 – Analyze and Solve Percent Problems
4		4	
5		5	
6		6	

Marking Period 3		Marking Period 4	
Topic/Unit Topic/Unit (use assessments as breaks if you divide a Unit) Identify what content will comprise each major assessment.		Topic/Unit Topic/Unit (use assessments as breaks if you divide a Unit) Identify what content will comprise each major assessment.	
1	Topic 8 – Solve Problems Involving Geometry	1	8 th Grade Topic 2- Analyze and Solve Linear Equations
2	8 th Grade Topic 7 – Understand and Apply the Pythagorean Theorem	2	8 th Grade Topic 3 – Use Functions to Model Relationships
3	Topic 7 – Probability	3	8 th Grade Topic 5 – Analyze and Solve Systems of Linear Equations
4	Topic 6 – Use Sampling to Draw Inferences about Populations	4	
5		5	
6		6	

Core Instructional Resources and Materials (Including Varied Levels of Text)

- Envision math 2.0 Authors: R Q Berry, III, Z Champagne, E Milou, J F Schielack, J A Wray, R I Charles, and F Fennell. Copyright 2017 by Pearson Education.
Adoption includes a workbook for each student as well as eBook access, Teacher Edition textbook and eBook access, and Savvas Easybridge website.
- **PLEASE NOTE:** Any resource or material that falls outside of the approved lists included within the curriculum document must have administrative approval prior to use. These resources or materials must also be noted in the teacher's lesson plan after approval is granted.

Topic # 1A (1.1 - 1.2, 1.6-1.10)	Duration (20 Blocks) *Note: one block is one day
Topic: Understand Ration Numbers, Multiply and Divide Integers and Rational Numbers	
Alignment to NJSLS (include practice standards)	
7.NS.A.1a: Describe situations in which opposite quantities combine to make 0. For example, in the first round of a game, Maria scored 20 points. In the second round of the same game, she lost 20 points. What is her score at the end of the second round? 7.NS.A.2a: Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers. Interpret products of rational numbers by describing real-world contexts. 7.NS.A.2b: Understand that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. If p and q are integers, then $-(p/q) = (-p)/q = p/(-q)$. Interpret quotients of rational numbers by describing real world contexts. 7.NS.A.2c: Apply properties of operations as strategies to multiply and divide rational numbers. 7.NS.A.2d: Convert a rational number to a decimal using long division; know that the decimal form of a rational number terminates in 0s or eventually repeats. 7.NS.A.3: Solve real-world and mathematical problems involving the four operations with rational numbers. MP.1: Make sense of problems and persevere in solving them MP.2: Reason abstractly and quantitatively MP.3: Construct viable arguments and critique the reasoning of others MP.4: Model with mathematics MP.6: Attend to precision MP.7: Look for and make use of structure MP.8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLS 9)	
Interdisciplinary Standards	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	

- Students may solve multi-step real-life problems posed with positive and negative rational numbers in any form related to the relationship between altitude and the temperature above sea level.

Learning Objectives and Activities

- SWBAT answer the following questions:
 - How are integers and their opposites related?
 - How are rational numbers written as decimals?
 - How do the signs of factors affect the product?
 - How is multiplying rational numbers like multiplying integers?
 - How does dividing integers relate to multiplying integers?
 - How do you decide which rational number operations to use to solve a problem?
- SWBAT demonstrate understanding of the following:
 - Connect opposites to the concept of absolute value.
 - Combine opposite quantities to make 0 in real-world context and represent the changes on a number line.
 - Describe how to convert a rational number to a decimal using long division
 - Explain in oral or written language why the decimal is either a terminating or repeating decimal.
 - Explain in oral or written language why a divisor cannot be zero and why division of integers results in a rational number.
 - Model the multiplication of signed numbers using real-world contexts
 - Multiply integers with same signs and integers with different signs
 - Understand that the signs of the factors affect the product.
 - Apply operations with rational numbers to real-world contexts.
 - Make comparisons between multiplying integers and rational numbers.
 - Explain in oral or written language sign rules when multiplying rational numbers.
 - Understand the relationship between multiplication and division to divide integers.
 - Determine which quotients of integers are equivalent.
 - Explain in oral or written language the relationship between multiplication and division when dividing integers.
 - Understand how the signs of integers in a multiplication sentence relate to the signs in a related division problem.
 - Extend knowledge about multiplying rational numbers and dividing integers to division of rational numbers
 - Model the multiplication and division of rational numbers using real-world contexts.
 - Explain in oral or written language the relationship between multiplication and division when dividing rational numbers.
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - P r a c t i c e a n d P r o b l e m S o l v i n g

- Differentiated Strategies
- Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
- Desmos activities

- Alternative
 - Desmos
 - IXL
 - Kahoot
 - Quizizz
 - Blooket
 - Projects
- Summative
 - Lesson quizzes
 - Mid Topic Assessments
 - End of Topic Assessments
 - Projects
- Formative
 - Classwork
 - Homework
 - Practice and Problem Solving
 - Additional Practice
 - IXL
 - Teacher Observation

Topic # 1B	Duration (10 Blocks) *Note: one block is one day
Topic: Add/Subtract Integers and Rational Numbers	
Alignment to NJSLS (include practice standards)	
7.NS.A.1: Understand $p + q$ as the number located a distance q from p, in the positive or negative direction depending on whether q is positive or negative. Show that a number and its opposite have a sum of 0 (are additive inverses). Interpret sums of rational numbers by describing real-world contexts. 7.NS.A.1c: Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Show that the distance between two rational numbers on the number line is the absolute value of their difference and apply this principle in real-world contexts. 7.NS.A.1d: Apply properties of operations as strategies to add and subtract rational numbers. MP.1: Make sense of problems and persevere in solving them MP.2: Reason abstractly and quantitatively MP.3: Construct viable arguments and critique the reasoning of others MP.4: Model with mathematics MP.6: Attend to precision MP.7: Look for and make use of structure MP.8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLS 9)	
Interdisciplinary Standards	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	
Students may solve multi-step real-life problems posed with positive and negative rational numbers in any form related to the relationship between altitude and the temperature above sea level.	
Learning Objectives and Activities	
- SWBAT answer the following questions: - How do you use what you know about absolute value to add integers? - How is subtracting integers related to adding integers?	

- How are adding and subtracting integers related to adding and subtracting other rational numbers?
- SWBAT demonstrate understanding of the following:
 - Add integers with same signs or different signs
 - Identify additive inverses and opposite integers using models such as number lines.
 - Apply the additive inverse property to subtract integers.
 - Recognize situations when the quantities result in a zero pair.
 - Extend knowledge of integers to evaluate differences between integers with the same or different signs.
 - Extend knowledge of positive and negative rational numbers to adding and subtracting with rational numbers.
 - Use number lines to model addition and subtraction as a movement or distance between rational numbers.
 - Explain in oral or written language sign rules when combining rational numbers.
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
 - Differentiated Strategies
 - Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
 - Desmos activities

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Desmos
 - IXL
 - Kahoot
 - Quizizz
 - Blooket
 - Projects
- Summative
 - Lesson quizzes
 - Mid Topic Assessments
 - End of Topic Assessments
 - Projects
- Formative
 - Classwork
 - Homework

- Practice and Problem Solving
- Additional Practice
- IXL
- Teacher Observation

Topic # 4	Duration (15 Blocks) *Note: one block is one day
Topic: Generate Equivalent Expressions	
Alignment to NJSLS (include practice standards)	
7.EE.A.1: Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients. 7.EE.A.2: Understand that rewriting an expression in different forms in a problem context can shed light on the problem and how the quantities in it are related. 7.EE.B.3 Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. 7.EE.B.4: Use variables to represent quantities in a real-world or mathematical problem and construct simple equations and inequalities to solve problems by reasoning about the quantities. MP.1: Make sense of problems and persevere in solving them MP.2: Reason abstractly and quantitatively MP.3: Construct viable arguments and critique the reasoning of others MP.4: Model with mathematics MP.6: Attend to precision MP.7: Look for and make use of structure MP.8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLS 9)	
● 8.1.2.DA.4: Make predictions based on data using charts or graphs.	
Interdisciplinary Standards	
● NJSLSA.SL1: Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively.	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	
Learning Objectives and Activities	

- SWBAT answer the following questions:
 - How can algebraic expressions be used to represent and solve problems?
 - What are equivalent expressions?
 - How are properties of operations used to simplify expressions?
 - How does the value of an expression change when it is expanded?
 - How does the Distributive Property relate to factoring expressions?
 - How can properties of operations be used to add expressions?
 - How can properties of operations be used to subtract expressions?
 - How can writing equivalent expressions show how quantities are related?
- SWBAT demonstrate understanding of the following:
 - Understand and write Algebraic Expressions
 - Understand how variables are used to represent unknown values in problems
 - Use Algebraic expressions to solve real world problems
 - Identify equivalent expressions
 - Write equivalent expressions using properties of operations and combining like terms
 - Identify like terms and use the combination of like terms to solve a problem
 - Develop fluency simplifying expressions
 - Apply Distributive Property to expand linear equations
 - Use models to solidify understanding of Distributive Property
 - Find common factors of linear expressions using Distributive Property
 - Recognize factoring is the opposite of expanding expressions
 - Use Commutative and Associative Properties to add expressions with rational coefficients
 - Apply understanding of adding expressions to solve real-world problems
 - Identify the similarity between subtracting integers and subtracting linear expressions
 - Simplify linear expressions using subtraction
 - Subtract linear expressions by applying properties of operations
 - Write equivalent expressions by combining like terms, using the Distributive Property, and performing other mathematical operations
 - Understand that equivalent expressions can be created by simplifying or expanding the original expression
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
 - D i f f e r e n t i a t e d S t r a t e g i e s

- Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
- Desmos activities

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 - IXL
 - Teacher Observation

Topic # 5	Duration (15 Blocks) *Note: one block is one day
Topic: Solve Problems Using Equations and Inequalities	
Alignment to NJSLs (include practice standards)	
7.EE.B.3: Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. 7.EE.B.4: Use variables to represent quantities in a real-world or mathematical problem and construct simple equations and inequalities to solve problems by reasoning about the quantities. 7.EE.B.4a: Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p, q, and r are specific rational numbers. Solve equations of these forms with accuracy and efficiency. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. 7.EE.B.4b: Solve word problems leading to inequalities of the form $px + q > r$ or $px + q < r$, where p, q, and r are specific rational numbers. Graph the solution set of the inequality and interpret it in the context of the problem. MP.1: Make sense of problems and persevere in solving them MP.2: Reason abstractly and quantitatively MP.3: Construct viable arguments and critique the reasoning of others MP.4: Model with mathematics MP.6: Attend to precision MP.7: Look for and make use of structure	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
8.1.2.DA.4: Make predictions based on data using charts or graphs.	
Interdisciplinary Standards	
NJSLSA.SL1: Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively.	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	
Learning Objectives and Activities	

- SWBAT answer the following questions:
 - How does an equation show the relationship between variables and other quantities in a situation?
 - How is solving a two-step equation similar to solving a one-step equation?
 - How does the Distributive Property help you solve equations?
 - How is solving inequalities with addition and subtraction similar to and different from solving equations with addition and subtraction?
 - How is solving inequalities with multiplication and division similar to and different from solving equations with multiplication and division?
 - How is solving a two-step inequality similar to and different from solving a two-step equation?
 - How is solving a multi-step inequality similar to and different from solving a multistep equation?
- SWBAT demonstrate understanding of the following:
 - Analyze word problems to write two-step equations
 - Interpret quantities in equations
 - Relate the structure of an equation, the properties of operations and the quantities of an equation
 - Compare algebraic and arithmetic solutions
 - Apply mathematical properties to solve for unknown variable in an equation
 - Solve word problems using bar diagrams
 - Use the Distributive Property to solve equations with integers and rational numbers
 - Use Distributive Property to separate a product into two terms to solve an equation
 - Use Distributive Property to Solve Real World problems
 - Using the Addition and Subtraction Properties of Inequality, solve inequality problems
 - Graph the solution sets of inequalities on a number line
 - Write and solve inequalities by using Multiplication and Division Properties of Inequality
 - Graph solutions of inequalities on the number line
 - Solve an inequality by multiplying or dividing by a negative rational number
 - Write a two-step inequality to solve a problem
 - Interpret the solutions of inequalities
 - Apply knowledge of solving two-step inequalities to solving multi-step inequalities
 - Explore the relationship between two-step inequalities and multi-step inequalities
 - Apply Distributive Property to simplify and solve multistep equations
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction

- Practice and Problem Solving
- Differentiated Strategies
- Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
- Desmos activities

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Desmos
 - IXL
 - Kahoot
 - Quizizz
 - Blooket
 - Projects
- Summative
 - Lesson quizzes
 - Mid Topic Assessments
 - End of Topic Assessments
 - Projects
- Formative
 - Classwork
 - Homework
 - Practice and Problem Solving
 - Additional Practice
 - IXL
 - Teacher Observation

Topic # 2	Duration (15 Blocks) *Note: one block is one day
Topic: Analyze and Use Proportional Relationships	
Alignment to NJSLs (include practice standards)	
7.RP.A.1: Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units. 7.RP.A.2: Recognize and represent proportional relationships between quantities. 7.RP.A.2a: Decide whether two quantities are in a proportional relationship, e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line through the origin. 7.RP.A.2b: Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships. 7.RP.A.2c: Represent proportional relationships by equations. 7.RP.A.2d: Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points (0, 0) and (1, r) where r is the unit rate 7.RP.A.3: Use proportional relationships to solve multistep ratio and percent problems. MP.1: Make sense of problems and persevere in solving them MP.2: Reason abstractly and quantitatively MP.3: Construct viable arguments and critique the reasoning of others MP.4: Model with mathematics MP.6: Attend to precision MP.7: Look for and make use of structure MP.8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
● 8.1.2.AP.4: Break down a task into a sequence of steps.	
Interdisciplinary Standards	
● NJSLSA.SL4: Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	

Learning Objectives and Activities

- SWBAT answer the following questions:
 - How are ratios, rates, and unit rates used to solve problems?
 - Why is it useful to write a ratio of fractions as a unit rate?
 - How are proportional quantities described by equivalent ratios?
 - How can you represent a proportional relationship with an equation?
 - What does the graph of a proportional relationship look like?
 - How can proportional reasoning help solve a problem?
- SWBAT demonstrate understanding of the following:
 - Explain the concept of unit rate
 - Compare unit rates of one or more real world problems
 - Use ratios to compare relationships between two quantities
 - Use mathematics modeling to represent a problem situation to propose a solution
 - Compute unit rates with ratios that contain one or two fractions
 - Use unit rates to make comparisons and solve problems
 - Look for equivalent ratios to identify proportional relationships
 - Use proportions to solve problems
 - Find the constant of proportionality in a proportional relationship
 - Write and use equations that represent situations involving proportional quantities
 - Recognize the graph of a proportional relationship as a straight line through the origin
 - Interpret points on a graph of a proportional relationship
 - Identify the constant of proportionality on the graph
 - Interpret a point on a graph of proportional relationship
 - Explain whether a situation represents a proportional relationship
 - Use representations to find entry points into problems
 - Explain whether a situation represents a proportional relationship
 - Distinguish between proportional and nonproportional relationships
 - Use reasoning to solve problems
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction

- Practice and Problem Solving
- Differentiated Strategies
- Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
- Desmos activities

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 - Teacher Observation

Topic # 3	Duration (15 Blocks) *Note: one block is one day
Topic: Analyze and Solve Percent Problems	
Alignment to NJSLS (include practice standards)	
7.RP.A.2c: Represent proportional relationships by equations. 7.RP.A.3: Use proportional relationships to solve multistep ratio and percent problems. MP.1: Make sense of problems and persevere in solving them MP.2: Reason abstractly and quantitatively MP.3: Construct viable arguments and critique the reasoning of others MP.4: Model with mathematics MP.6: Attend to precision MP.7: Look for and make use of structure MP.8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLS 9)	
● 8.1.2.AP.4: Break down a task into a sequence of steps.	
Interdisciplinary Standards	
● NJSLSA.SL4: Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	
Learning Objectives and Activities	
● SWBAT answer the following questions: ○ How do percents show the relationship between quantities? ○ How does proportional reasoning relate to percent? ○ How are percent problems related to proportional reasoning? ○ How is finding percent error similar to finding percent change? ○ How are the concepts of percent markup and percent markdown related to percent equation? ○ How does simple interest show proportional reasoning and relate to the percent equation? ● SWBAT demonstrate understanding of the following:	

- Represent percent as a ratio, including those greater than 100% and less than 1%
- Understand that equivalent ratios can be used to calculate percent
- Analyze percent of numbers in a real-world context
- Write a proportion to model a percent situation
- Use percent proportion to find an unknown part, whole or percent
- Use bar diagrams and proportions to solve percent problems
- Use proportional reasoning to write percent equations
- Write and use percent equations to solve problems involving proportional relationships
- Calculate percent change and percent error in real world applications
- Modify percent equation to find percent change
- Interpret the results of a percent equation in a real life scenario
- Solve real-world problems involving percent change and percent error
- Use mathematical modeling to represent a problem situation and to propose a solution.
- Relate percent change to percent markup
- Relate percent change to percent markdown
- Solve real world problems involving markup/markdown
- Understand what simple interest is
- Calculate simple interest
- Apply percent equation to solve simple interest problems.
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
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Topic # 8	Duration (12 Blocks) *Note: one block is one day
Topic: Solve Problems Involving Geometry	
Alignment to NJSLs (include practice standards)	
7.G.A.1: Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale. 7.G.A.2: Draw (with technology, with ruler and protractor, as well as freehand) geometric shapes with given conditions. Focus on constructing triangles from three measures of angles or sides, noticing when the conditions determine a unique triangle, more than one triangle, or no triangle. 7.G.A.3: Describe the two-dimensional figures that result from slicing three-dimensional figures, as in plane sections of right rectangular prisms and right rectangular pyramids. 7.G.B.4: Know the formulas for the area and circumference of a circle and use them to solve problems; give an informal derivation of the relationship between the circumference and area of a circle. 7.G.B.5: Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle in a figure. 7.EE.B.3: Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. 7.EE.B.4a: Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p, q and r are specific rational numbers. Solve equations of these forms with accuracy and efficiency. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. MP.1: Make sense of problems and persevere in solving them MP.2: Reason abstractly and quantitatively MP.3: Construct viable arguments and critique the reasoning of others MP.4: Model with mathematics MP.5: Use appropriate tools strategically MP.6: Attend to precision MP.7: Look for and make use of structure MP.8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
8.1.8.DA.1: Organize and transform data collected using computational tools to make it usable for a specific purpose.	

Interdisciplinary Standards	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	
Students may solve real-world problems involving area, surface area, and volume related to deforestation and increasing livestock farming as key contributors to climate change.	
Learning Objectives and Activities	
SWBAT answer the following questions: - How do scale drawings and actual measurements represent proportional relationships? - How can a shape that meets given conditions be drawn? - How can you determine when it is possible to draw a triangle given certain conditions? - How are angles formed by intersecting lines related? - How is the circumference of a circle related to the length of its diameter? - How can the area formula for a circle be used to solve problems? - How do the faces of a three-dimensional figure determine the two-dimensional shapes created by slicing the figure? - How is finding the area of composite 2-dimensional figures similar to finding the surface area of a 3-dimensional figure? - How does the formula for volume of a prism help you understand what the volume of a prism means? SWBAT demonstrate understanding of the following: - Using a scale drawing as a representation of actual lengths and area. - Sketch quadrilaterals with given conditions. - Name and classify quadrilaterals according to their properties. - Construct triangles with given conditions. - Conclude whether a triangle is formed and what type of triangle it is. - Calculate the measures of angles by using angle relationships. - Calculate circumference, radius, or diameter of a circle. - Recognize the relationship between the circumference and diameter of a circle and pi. - Find the area of a circle. - Use the area to find the radius and diameter. - Solve problems involving the area of a circle. - Use mathematical modeling to represent a problem situation and propose a solution. - Test and verify the appropriateness of their mathematical models. - Describe cross sections of right rectangular prisms and pyramids. - Solve problems involving cross sections. - Find the surface area of two-dimensional composite shapes. - Find the surface area of three-dimensional composite shapes.	

- Calculate the volume of various three-dimensional figures.
 - Solve problems involving the volume of three-dimensional figures.
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
 - Differentiated Strategies
 - Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
 - Desmos activities

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Desmos
 - IXL
 - Kahoot
 - Quizizz
 - Blooket
 - Projects
- Summative
 - Lesson quizzes
 - Mid Topic Assessment
 - End of Topic Assessments
 - Projects
- Formative
 - Classwork
 - Homework
 - Practice and Problem Solving
 - Additional Practice
 - IXL
 - Teacher Observation
 - Desmos Activities

8 th Grade Topic # 7	Duration (10 Blocks) *Note: one block is one day
Topic: Understand and Apply the Pythagorean Theorem	
Alignment to NJSLs (include practice standards)	
8.G.B.6: Explain a proof of the Pythagorean Theorem and its converse. 8.G.B.7: Apply the Pythagorean Theorem to determine unknown side lengths in right triangles in real-world and mathematical problems in two and three dimensions. 8.G.B.8: Apply the Pythagorean Theorem to find the distance between two points in a coordinate system. MP1: Make sense of problems and persevere in solving them MP2: Reason abstractly and quantitatively MP3: Construct viable arguments and critique the reasoning of others MP4: Model with mathematics MP7: Look for and make use of structure MP8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
Interdisciplinary Standards	
NJSLSA.SL1: Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively. NJSLSA.SL4: Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	
Learning Objectives and Activities	
• SWBAT answer the following questions: ○ How does the Pythagorean Theorem relate the side lengths of a right triangle?	

- How can you determine whether a triangle is a right triangle?
- What types of problems can be solved using the Pythagorean Theorem?
- SWBAT demonstrate understanding of the following:
 - Understand the proof of the Pythagorean Theorem
 - The Pythagorean Theorem can be used to find an unknown side length of a right triangle.
 - If a triangle has side lengths a , b and c such that the sum of the squares of the two legs equals the square of the hypotenuse.
 - Understand and apply the Converse of the Pythagorean Theorem.
 - Use the Pythagorean Theorem and its converse to solve problems.
 - The Pythagorean Theorem can be used to find the distance between any two points on a coordinate plane by drawing a line to connect the two points and by using the line as the hypotenuse of a right triangle.
- Learning Activities
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
 - Differentiated Strategies
 - Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
 - Desmos activities

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Desmos
 - IXL
 - Kahoot
 - Quizizz
 - Blooket
 - Projects
- Summative
 - Lesson quizzes
 - Mid Topic Assessments
 - End of Topic Assessments
 - Projects
- Formative

- Classwork
- Homework
- Practice and Problem Solving
- Additional Practice
- IXL
- Teacher Observation
- Desmos activities

Topic # 7	Duration (12 Blocks) *Note: one block is one day
Topic: Probability	
Alignment to NJSLs (include practice standards)	
7.SP.C.5: Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event, a probability around $\frac{1}{2}$ indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event. 7.SP.C.6: Approximate the probability of a chance event by collecting data on the chance process that produces it and observing its long-run relative frequency and predict the approximate relative frequency given the probability. 7.SP.C.7: Develop a probability model and use it to find probabilities of events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy. 7.SP.C.7a: Develop a uniform probability model by assigning equal probability to all outcomes and use the model to determine probabilities of events. 7.SP.C.7b: Develop a probability model (which may not be uniform) by observing frequencies in data generated from a chance process. 7.SP.C.8a: Understand that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space for which the compound event occurs. 7.SP.C.8b: Represent sample spaces for compound events using methods such as organized lists, tables and tree diagrams. For an event described in everyday language (e.g., “rolling double sixes”), identify the outcomes in the sample space which compose the event. 7.SP.C.8c: Design and use a simulation to generate frequencies for compound events. 7.EE.B.3: Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. 7.RP.A.2c: Represent proportional relationships by equations. MP1: Make sense of problems and persevere in solving them MP2: Reason abstractly and quantitatively MP3: Construct viable arguments and critique the reasoning of others MP4: Model with mathematics MP6: Attend to precision MP7: Look for and make use of structure MP8: Look for and express regularity in repeated reasoning	

Technology, Career Awareness, and 21st Century Standards (NJSLS 9)

9.4.12.CI.1: Demonstrate the ability to reflect, analyze, and use creative skills and ideas.

Interdisciplinary Standards

NJSLSA.SL1: Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively.

NJSLSA.SL4: Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.

Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)

Learning Objectives and Activities

- SWBAT answer the following questions:
 - What is probability?
 - How can the probability of an event help make predictions?
 - How is experimental probability similar to and different from theoretical probability?
 - How can a model be used to find the probability of an event?
 - How can all the possible outcomes, or sample space, of a compound event be represented?
 - How can a model help find the probability of a compound event?
 - How can you use simulations to determine the probability of events?
- SWBAT demonstrate understanding of the following:
 - Use probability to describe the likelihood that an event will occur.
 - Apply knowledge or probability to relate probability to mathematical fairness.
 - Use theoretical probability to predict outcomes.
 - Identify and describe events.
 - Identify Theoretical probability.
 - Use prior knowledge to compare Theoretical Probability to Experimental Probability.
 - Make predictions using Experimental Probability.
 - Develop probability models.
 - Use Probability model to evaluate a situation.
 - Use Probability model to make an estimate.
 - Represent a sample space for a compound event by using a tree diagram, a table or an organized list.
 - Use a table, tree diagram or a table to organize information about a compound event.
 - Determine the probability of a compound event.

- Use a variety of tools (spinner, coin, number cube, or random number generator) to simulate a compound event.
- Model a real-world situation involving a compound event.
- Predict the outcome of a real-world situation involving a compound event using a simulation.
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
 - Differentiated Strategies
 - Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
 - Desmos Activities

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Desmos
 - IXL
 - Kahoot
 - Quizizz
 - Blooket
 - Projects
- Summative
 - Lesson quizzes
 - Mid topic Assessments
 - End of Topic Assessments
 - Projects
- Formative
 - Classwork
 - Homework
 - Practice and Problem Solving
 - Additional Practice
 - IXL
 - Teacher Observation

Topic # 6	Duration (8 Blocks) *Note: one block is one day
Topic: Use Sampling to Draw Inferences About Populations	
Alignment to NJSLs (include practice standards)	
7.SP.A.1: Understand that statistics can be used to gain information about a population by examining a sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population. Understand that random sampling tends to produce representative samples and support valid inferences. 7.SP.A.2: Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated samples) of the same size to gauge the variation in estimates or predictions 7.SP.B.3: Informally assess the degree of visual overlap of two numerical data distributions with similar variabilities, measuring the difference between the centers by expressing it as a multiple of a measure of variability. 7.SP.B.4: Use measures of center and measures of variability for numerical data from random samples to draw informal comparative inferences about two populations. 7.RP.A.2c: Represent proportional relationships by equations. 7.EE.B.3: Solve multi-step real-life and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate; and assess the reasonableness of answers using mental computation and estimation strategies. MP1: Make sense of problems and persevere in solving them MP2: Reason abstractly and quantitatively MP3: Construct viable arguments and critique the reasoning of others MP4: Model with mathematics MP6: Attend to precision MP8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
9.4.12.CI.1: Demonstrate the ability to reflect, analyze, and use creative skills and ideas.	
Interdisciplinary Standards	
NJSLSA.SL1: Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively.	

NJSLSA.SL4: Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.

Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)

- Students may solve multi-step real-life problems posed with positive and negative rational numbers in any form related to the relationship between altitude and the temperature above sea level.

Learning Objectives and Activities

- SWBAT answer the following questions:
 - How can you determine a representative sample of a population?
 - How can inferences be drawn about a population from data gathered from sample?
 - How can data displays be used to compare populations?
- SWBAT demonstrate understanding of the following:
 - Understand the difference between a population and a sample using real world scenarios.
 - Establish whether a sample is representative of a population.
 - Generate random samples that represent the entire population.
 - Understand the differences between qualitative and quantitative inferences.
 - Make qualitative and quantitative inferences from a random sample data set.
 - Make estimates about a population based on a sample data set.
 - Assess whether inferences are valid.
 - Use box plots to compare and make inferences about populations.
 - Use the median and interquartile range of datasets to informally compare and make inferences about two populations.
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
 - Differentiated Strategies
 - Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
 - Desmos Activities
- **Note:** Lesson 6.4 will be a Flex lesson

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Desmos
 - IXL
 - Kahoot
 - Quizizz
 - Blooket
 - Projects
- Summative
 - Lesson quizzes
 - Mid Topic Assessments
 - End of Topic Assessments
 - Projects
- Formative
 - Classwork
 - Homework
 - Practice and Problem Solving
 - Additional Practice
 - IXL
 - Teacher Observation

8 th Grade Topic # 2	Duration (15 Blocks) *Note: one block is one day
Topic: Analyze and Solve Linear Equations	
Alignment to NJSLs (include practice standards)	
8.EE.B.5: Graph proportional relationships, interpreting the unit rate as the slope of the graph. Compare two different proportional relationships represented in different ways. 8.EE.B.6: Use similar triangles to explain why the slope m is the same between any two distinct points on a non-vertical line in the coordinate plane; derive the equation $y = mx$ for a line through the origin and the equation $y = mx + b$ for a line intercepting the vertical axis at b. 8.EE.C.7a: Give examples of linear equations in one variable with one solution, infinitely many solutions, or no solutions. Show which of these possibilities is the case by successively transforming the given equation into simpler forms, until an equivalent equation of the form $x = a$, $a = a$, or $a = b$ results (where a and b are different numbers). 8.EE.C.7b: Solve linear equations with rational number coefficients, including equations whose solutions require expanding expressions using the distributive property and collecting like terms. MP1: Make sense of problems and persevere in solving them MP2: Reason abstractly and quantitatively MP3: Construct viable arguments and critique the reasoning of others MP4: Model with mathematics MP6: Attend to precision MP7: Look for and make use of structure MP8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
8.1.8.AP.2: Create clearly named variables that represent different data types and perform operations on their values. 9.4.12.CI.1: Demonstrate the ability to reflect, analyze, and use creative skills and ideas.	
Interdisciplinary Standards	
NJSLSA.SL1: Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively. NJSLSA.SL4: Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.	

Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)

Learning Objectives and Activities

- **SWBAT** answer the following questions:
 - How do you use inverse operations to solve equations with variables on both sides?
 - Will a one-variable equation always have only one solution?
 - What is slope?
 - How does slope relate to the equation for a proportional relationship?
 - What is the y-intercept and what does it indicate?
 - What is the equation of a line for a nonproportional relationship?
- **SWBAT** demonstrate understanding of the following:
 - Solve equations with like terms on both sides of the equation
 - Make sense of scenarios and represent them with equations
 - Determine the number of solutions to an equation
 - Find the slope of a line using different strategies
 - Interpret a slope in context and relate it to the steepness of a graph
 - Understand how the constant of proportionality and the slope relate to a linear equation
 - Write a linear equation in the form $y = mx$ when the slope is given
 - Graph a linear equation in the form $y = mx$
 - Interpret and extend the table or graph of a linear relationship to find its y-intercept
 - Analyze graphs in context to determine and explain the meaning of the y-intercept
 - Graph a line from an equation in the form $y = mx + b$
 - Write an equation that represents the given graph of a line
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
 - Differentiated Strategies
 - Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
 - Desmos activities
- Lessons 2.1, 2.3 and 2.5 are flex lessons as they are introduced earlier in the year.

Benchmark, Formative, Summative and Alternative Assessments
● Alternative ○ Desmos ○ IXL ○ Kahoot ○ Quizizz ○ Blooket ○ Projects ● Summative ○ Lesson quizzes ○ Mid Topic Assessments ○ End of Topic Assessments ○ Projects ● Formative ○ Classwork ○ Homework ○ Practice and Problem Solving ○ Additional Practice ○ IXL ○ Teacher Observation

- Alternative

- Desmos
- IXL
- Kahoot
- Quizizz
- Blooket
- Projects

- Summative

- Lesson quizzes
- Mid Topic Assessments
- End of Topic Assessments
- Projects

- Formative

- Classwork
- Homework
- Practice and Problem Solving
- Additional Practice
- IXL
- Teacher Observation

8 th Grade Topic # 3	Duration (12 Blocks) *Note: one block is one day
Topic: Use Functions to model Relationships	
Alignment to NJSLs (include practice standards)	
8.F.A.1: Understand that a function is a rule that assigns to each input exactly one output. The graph of a function is the set of ordered pairs consisting of an input and the corresponding output. 8.F.A.2: Compare properties (e.g. rate of change, intercepts, domain and range) of two functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions). For example, given a linear function represented by a table of values and a linear function represented by an algebraic expression, determine which function has the greater rate of change. 8.F.A.3: Interpret the equation $y = mx + b$ as defining a linear function, whose graph is a straight line; give examples of functions that are not linear. For example, the function $A = s^2$ giving the area of a square as a function of its side length is not linear because its graph contains the points (1,1), (2,4) and (3,9), which are not on a straight line. 8.F.B.4: Construct a function to model a linear relationship between two quantities. Determine the rate of change and initial value of the function from a description of a relationship or from two (x, y) values, including reading these from a table or from a graph. Interpret the rate of change and initial value of a linear function in terms of the situation it models, and in terms of its graph or a table of values. 8.F.B.5: Describe qualitatively the functional relationship between two quantities by analyzing a graph (e.g., where the function is increasing or decreasing, linear or nonlinear). Sketch a graph that exhibits the qualitative features of a function that has been described verbally. MP1: Make sense of problems and persevere in solving them MP2: Reason abstractly and quantitatively MP3: Construct viable arguments and critique the reasoning of others MP4: Model with mathematics MP5: Use appropriate tools strategically MP7: Look for and make use of structure MP8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
Interdisciplinary Standards	

NJSLSA.SL1: Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively.

NJSLSA.SL4: Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.

Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)

Learning Objectives and Activities

- SWBAT answer the following questions:
 - When is a relation a function?
 - When are different representations a function?
 - How can you compare two functions?
 - How can you use a function to represent a linear relationship?
 - How does a qualitative graph describe the relationship between quantities?
 - How does the sketch of a graph of a function describe its behavior?
- SWBAT demonstrate understanding of the following:
 - Identify whether a relation is a function.
 - Interpret a function
 - Identify functions in different representations: equations, tables and graphs
 - Identify linear and nonlinear functions in different representations
 - Compare properties of linear functions in different representations
 - Compare properties of linear and nonlinear functions in different representations
 - Use mathematical modeling to represent a problem situation and to propose a solution
 - Test and verify the appropriateness of their math models
 - Explain why the results from their mathematical models may not align exactly to the problem
 - Describe qualitatively the behavior of a function by analyzing its graph
 - Describe the graph of a function at each interval
 - Draw a qualitative graph of a function based of a verbal description
 - Analyze and interpret the sketch of a graph of a function
- Learning Activities
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning

- Formal Direct Instruction
- Practice and Problem Solving
- Differentiated Strategies
- Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
- Desmos activities

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Desmos
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 - Projects
- Summative
 - Lesson quizzes
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 - Projects
- Formative
 - Classwork
 - Homework
 - Practice and Problem Solving
 - Additional Practice
 - IXL
 - Teacher Observation

8 th Grade Topic # 5 (8 th Grade material)	Duration (15 Blocks) *Note: one block is one day
Topic: Analyze and Solve Systems of Linear Equations	
Alignment to NJSLS (include practice standards)	
8.EE.C.8a: Understand that solutions to a system of two linear equations in two variables correspond to points of intersection of their graphs, because points of intersection satisfy both equations simultaneously. 8.EE.C.8b: Solve systems of two linear equations in two variables using the substitution method and estimate solutions by graphing the equations. Solve simple cases by inspection. 8.EE.C.8c: Solve real-world and mathematical problems leading to two linear equations in two variables. MP1: Make sense of problems and persevere in solving them MP2: Reason abstractly and quantitatively MP3: Construct viable arguments and critique the reasoning of others MP4: Model with mathematics MP6: Attend to precision MP7: Look for and make use of structure MP8: Look for and express regularity in repeated reasoning	
Technology, Career Awareness, and 21st Century Standards (NJSLS 9)	
Interdisciplinary Standards	
NJSLSA.SL1: Prepare for and participate effectively in a range of conversations and collaborations with diverse partners, building on others' ideas and expressing their own clearly and persuasively. NJSLSA.SL4: Present information, findings, and supporting evidence such that listeners can follow the line of reasoning and the organization, development, and style are appropriate to task, purpose, and audience.	
Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)	
Learning Objectives and Activities	

- SWBAT answer the following questions:
 - How are slopes and y-intercepts related to the number of solutions of a system of linear equations?
 - How does the graph of a system of equations represent its solution?
 - When is substitution a useful method for solving systems of equations?
 - How are the properties of equality used to solve systems of linear equations?
- SWBAT demonstrate understanding of the following:
 - Examine graphs of linear systems of equations to determine the number of solutions, based on number of inspection points
 - Compare the equations in a linear system to look for a relationship between the number of solutions and the slopes and y-intercepts of equations
 - Create and examine graphs of linear systems of equations to determine the solution
 - Understand how substitution can be used to solve a linear system of equations
 - Apply this understanding to solve systems of equations with one solution, no solutions or infinitely many solutions
 - Understand how the process of elimination can be used to solve a system of linear equations with one solution, no solutions or infinitely many solutions
 - Apply this understanding to solve mathematical and real-world problems
- Learning Activities:
 - Problem-Based Learning
 - Solve and Discuss Problem
 - Look for Relationships
 - Visual Learning
 - Formal Direct Instruction
 - Practice and Problem Solving
 - Differentiated Strategies
 - Math Tools, Video Tutorials and Math Games: Online at [PearsonRealize.com](https://www.pearsonrealize.com)
 - Desmos activities

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Desmos
 - IXL
 - Kahoot
 - Quizizz
 - Blooket
 - Projects
- Summative
 - Lesson quizzes

- Mid Topic Assessments
 - End of Topic Assessments
 - Projects
- Formative
 - Classwork
 - Homework
 - Practice and Problem Solving
 - Additional Practice
 - IXL
 - Teacher Observation
 - Desmos Activities

Modifications (ML, Special Education, At Risk Students, Gifted & Talented, & 504 Plans)

ML

- Use visuals/anchor charts
- Introduce key vocabulary before lesson
- Teacher models reading aloud daily
- Guided notes

Supports for Students With IEPs:

- Allow extra time to complete assignments or tests
- Allow use of calculators
- Allow use of manipulatives
- Guided notes and/or scaffold outline for written assignments
- Work in a small group
- Allow answers to be given orally or dictated
- Use large print books, Braille, or e-books
- Follow all IEP modifications

At-Risk Students:

- Guided notes and/or scaffold outline for written assignments
- Introduce key vocabulary before lesson
- Work in a small group
- Lesson taught again using a differentiated approach
- Allow answers to be given orally or dictated
- Use visuals / anchor charts
- Leveled texts according to ability

Gifted and Talented:

- Create an enhanced set of introductory activities (e.g. advance organizers, concept maps, concept puzzles)
- Provide options, alternatives and choices to differentiate and broaden the curriculum
- Organize and offer flexible small group learning activities
- Provide whole group enrichment explorations
- Use center, stations, or contracts
- Organize integrated problem-solving simulations
- Propose interest-based extension activities
- Expose students to beyond level texts.

Supports for Students With 504 Plans:

- Follow all the 504 plan modifications
- Allow use of calculators

- Allow use of manipulatives
- Text to speech/audio recorded selections
- Amplification system as needed
- Leveled texts according to ability
- Fine motor skill stations embedded in rotation as needed
- Provide anchor charts with high frequency words and phonemic patterns